

START — Case / Real-time event

1 Step 1
Load case config
 f_0, R_f , fault type

2 Step 2
Forward Simulation
 $\{i_a, i_b, i_c\}(t)$

3 Step 3
Fixed OC Relay
 $t_{trip}^{fix}, I_{rms}(t)$

4 Step 4
Event Detection
 $\hat{t}_0$ via RMS jump

5 Step 5
Window Extraction
 $[t_0 - \delta_{pre}, t_c - \delta_{end}]$

6 Step 6
Two-Pass Estimator
 $\hat{\theta}$

7 Step 7
Sensitivity +
Confidence
 $\kappa_n(J), \gamma$

$\gamma \geq \gamma_{th}?$

No

Yes

Reject
Retain fixed settings
 I_p, TMS

Accept
Map $\hat{\theta} \rightarrow (I_p^*, TMS^*)$
Bounded update

Adaptive OC relay
 t_{trip}^{adapt}

9 Step 9
Metrics: RMSE, R^2, γ ,
 Δt_{trip} , security

10 Step 10
Save outputs: CSV,
log, plots

NEXT EVENT / CASE